

FINAL PROJECT REPORT

Smartphone Usage , Digital Addiction & Well-Being Analytics

A Power BI Dashboard and Report on Smartphone Usage, Sleep, Stress and Addiction

Submitted by

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1. Introduction

1.1. Project Overview

Smartphones are central to the daily life of students and young professionals, but heavy use is linked to poor sleep, higher stress and weaker academic or work performance. This project, **Digital Well-Being & Smartphone Addiction Insights**, analyses the smartphone usage of **7,500 users aged 18 to 35**. Each record covers daily and weekend screen time, social media, gaming and work/study hours, sleep, notifications, app opens, stress level, academic/work impact and addiction level. The data was profiled with SQL (MySQL) and modelled and visualised in **Microsoft Power BI**.

The Power BI file (smartphone_usage.pbix) has one data table and five pages: **KPI** (11 cards), **Visualizations** (exploratory charts and slicers), **Dashboard** (one-page summary), **Report 1** (behaviour analysis with a Decomposition Tree) and **Report 2** (well-being insights with a Key Influencers analysis).

1.2. Objectives

1. Load, profile and validate the dataset with SQL and assess its quality.
2. Summarise usage behaviour with KPIs (screen time, sleep, notifications, app opens).
3. Compare addicted and non-addicted users on screen time, social media, gaming and weekend use.
4. Study how screen time relates to sleep, stress and academic/work impact across gender, age group and addiction level.
5. Identify the strongest drivers of addiction and deliver an interactive dashboard and report for counsellors.

2. Project Initialization and Planning Phase

2.1. Define Problem Statement

Usage data is easy to collect but hard to interpret. It is not clear **which behaviours are linked to addiction**, whether addiction shows up in sleep or stress, or which groups need attention first. This project addresses the lack of a clear, interactive view that connects raw usage records to addiction outcomes.

Item	PS-1: Young user	PS-2: Well-being counsellor
I am	A student or young professional (18–35) using my phone for study, work and fun.	A counsellor responsible for a large group of students.
Trying to	Understand if my phone habits hurt my sleep, studies and stress.	Identify users at risk of addiction and plan support.
But	I cannot tell which habit (screen time, social media, gaming, notifications) matters.	Usage records are large tables that are hard to compare.
Because	There is no benchmark to compare myself against.	There is no single view linking usage to addiction, stress and sleep.
Which makes me feel	Anxious and unsure how to change.	Unable to prioritise interventions.

2.2. Project Proposal (Proposed Solution)

Parameter	Description
Solution	Profile the data with SQL, add calculated columns (Age Group, Addicted Status, High Screen Time User, User Count) and publish a five-page interactive Power BI dashboard and report with slicers, KPIs, a Decomposition Tree and Key Influencers.
Novelty	Combines usage, sleep, stress, academic/work impact and addiction level in one view and uses AI visuals to rank the drivers of addiction.
Social impact	Helps students, parents, colleges and employers spot unhealthy phone habits early.
Business model	Reporting template for colleges, corporate wellness programmes and digital well-being apps.
Scalability	One 16-column table; can be refreshed with new records or connected to a live database.

2.3. Initial Project Planning

#	Phase	Activities and output
1	Problem definition	Topic, problem statement and business questions.
2	Data collection	Obtain Smartphone_Usage.csv and load it into MySQL database "smartphone".
3	SQL profiling	Summary, null-check and group-by queries (smartphone_usage.sql).
4	Data preparation	Import into Power BI, set data types, create calculated columns.
5	Visuals, dashboard, report	Build cards, charts, slicers, Decomposition Tree, Key Influencers; assemble 5 pages (smartphone_usage.pbix).
6	Testing and documentation	Check filters and interactions; prepare this report.

3. Data Collection and Preprocessing Phase

3.1. Data Collection Plan and Raw Data Sources Identified

One structured dataset was used: **Smartphone_Usage.csv** (7,500 rows × 16 columns, about 0.6 MB), one record per user with unique transaction_id and user_id. It was stored in MySQL (database "smartphone", table "smartphone_usage") and imported into Power BI as one table. The columns are:

Column	Description	Column	Description
transaction_id, user_id	Unique record and user IDs	sleep_hours	Average nightly sleep (4.5–9.0 h)
age	Years (18–35)	notifications_per_day	Notifications received (20–250)
gender	Male, Female, Other	app_opens_per_day	App opens per day (15–180)
daily_screen_time_hours	Total daily screen time (3.0–12.0 h)	weekend_screen_time	Screen time per weekend day (3.58–14.88 h)
social_media_hours	Daily social media (0.5–6.0 h)	stress_level	Low, Medium, High
gaming_hours	Daily gaming (0–4 h)	academic_work_impact	Yes / No: phone use affects study or work
work_study_hours	Daily work/study use (0.5–6.0 h)	addiction_level	Mild, Moderate, Severe; blank if none
		addicted_label	1 = addicted, 0 = not addicted

3.2. Data Quality Report

The dataset is of high quality: no duplicates, no out-of-range values and only one column with blanks, and those blanks are meaningful.

Check	Finding	Action
Completeness	Only addiction_level has blanks: 819 of 7,500 (10.9%) ; the other 15 columns are complete.	No rows dropped.
Nature of blanks	All 819 belong to users with addicted_label = 0 (no addiction).	Kept as a "no level" group, not imputed.
Label consistency	Moderate (2,874) and Severe (2,434) are always addicted = 1; Mild (1,373) and blank are always 0.	No conflicts.
Duplicates / categories	0 duplicate rows; IDs 100% unique; no spelling variants in gender, stress, impact or addiction level.	None required.
Ranges / outliers	All values plausible; the IQR rule flags no outliers in screen time or sleep.	All values retained.
Logic	Weekend screen time $\geq$ daily average for every user (+1.74 h on average). But social media + gaming + work/study exceeds total screen time for 60.7% of users.	These columns are treated as separate measures and never added together.

3.3. Data Exploration and Preprocessing

Five queries in smartphone_usage.sql (Appendix 10.1) gave these results.

Q1: average daily screen time 7.50 h, average sleep 6.74 h, 34.1% High stress. **Q2:** only addiction_level has nulls (819).

Q3: stress levels are split almost evenly for both addicted (High 1,801, Low 1,782, Medium 1,725) and non-addicted users (759, 721, 712).

Q4: see the table.

Q5: users with and without academic/work impact have nearly identical screen time (7.49 vs 7.51 h), social media (3.28 vs 3.26 h) and gaming (2.01 vs 2.02 h).

addiction_level	Users	Avg daily screen time (h)	Avg sleep (h)
Mild	1,373	5.54	6.66
Moderate	2,874	8.35	6.76
Severe	2,434	8.60	6.78
NULL (none)	819	4.52	6.68

In Power BI the table was imported as "smartphone smartphone_usage" (columns shown with spaces), data types were checked, blank addiction levels were retained, and four calculated columns were added:

Calculated column	Purpose and logic
Age Group	Bands the age for slicers and charts ("18-35").
Addicted Status	Text version of addicted_label ("Addicted" or non-addicted); used by KPI filters and Key Influencers.
High Screen Time User	Flags users above a screen-time cut-off. [12 > 8 hours.]
User Count	Value 1 per record, so Sum(User Count) = number of users.

4. Data Visualization

4.1. Framing Business Questions

ID	Business question	Answered by
BQ1	What is the average daily screen time when categorized by gender?	Column chart comparing average daily screen time hours for Male, Other, and Female categories.
BQ2	How many users belong to each specific digital addiction level?	Horizontal bar chart comparing user counts across Moderate, Severe, Mild, and None levels.
BQ3	How does average daily screen time vary across different age groups?	Column chart tracking average daily screen time across groups: 30-39, 20-29, and Under 20.
BQ4	What is the distribution and proportion of addicted vs non-addicted users in the dataset?	Donut chart representing total tracked users (7.5K) divided into group 1 (29.23%) vs group 0 (70.77%).
BQ5	Is there a visible correlation or relationship between a user's daily screen time and their sleep hours?	High-density scatter plot distributing coordinates mapping sleep hours against daily screen time hours.
BQ6	How does average gaming time compare directly to social media usage hours?	Comparative bar chart contrasting the average of social media hours with the average of gaming hours.
BQ7	How can gender and addiction-level filters be used to view the dashboard dynamically?	Slicers – Gender and Addiction Level Shows the interactive filters available for Female, Male, Other and the addiction-level categories.
BQ8	How can the dashboard be filtered by age group?	Slicer – Age Group Shows the age-group filter available for 20–29, 30–39 and Under 20.

4.2. Developing Visualizations

The report has **5 pages** and **56 visual objects**. Page 1 (KPI) and page 2 (Visualizations) are described here; the Dashboard, Report 1 and Report 2 pages follow in Sections 5 and 6.

Page	Visuals
KPI	11 cards: total users, addicted users (filtered to Addicted Status = Addicted), and averages of age, daily and weekend screen time, social media, gaming, work/study hours, sleep, notifications and app opens.
Visualizations	Average daily screen time by gender (column) and by Age Group (column); Users by addiction level (bar); Addicted vs Non-Addicted Users (donut); Daily Screen Time vs Sleep Hours (scatter); Average Gaming vs Social Media (column); slicers for gender, Age Group and addiction level.



Figure 4.1: KPI Page

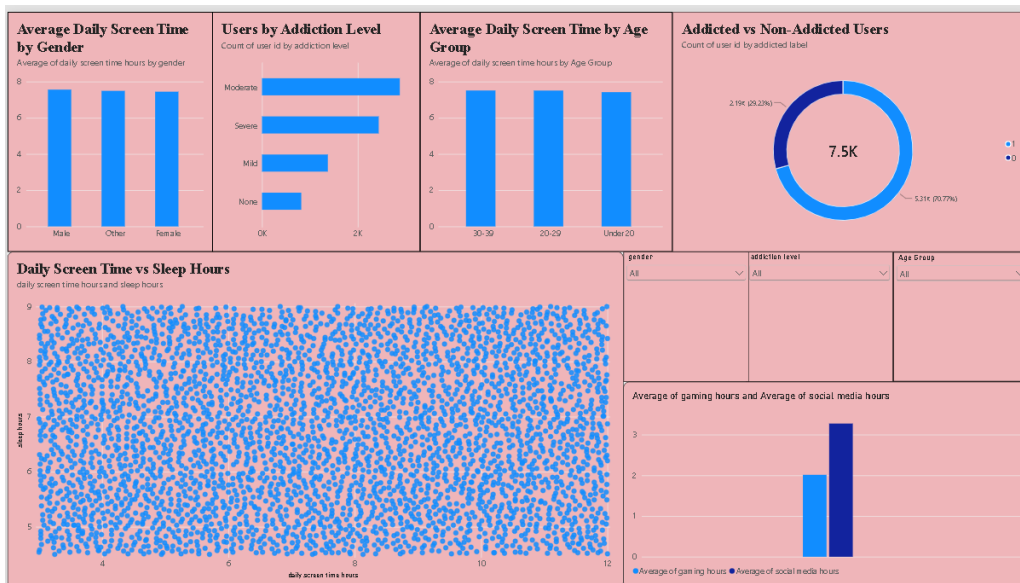


Figure 4.2: Visualization Page

5. Dashboard

5.1. Dashboard Design Screenshot with Insights

The **Dashboard** page ("Digital Well-Being & Addiction Insights Dashboard") has four cards (total users, addicted users, average weekend screen time, average sleep), an addiction-level slicer, and six charts: daily screen time by gender, users by addiction level (pie), sleep by addiction level, users by age group (pie), app opens by stress level, and weekend screen time by addiction status.

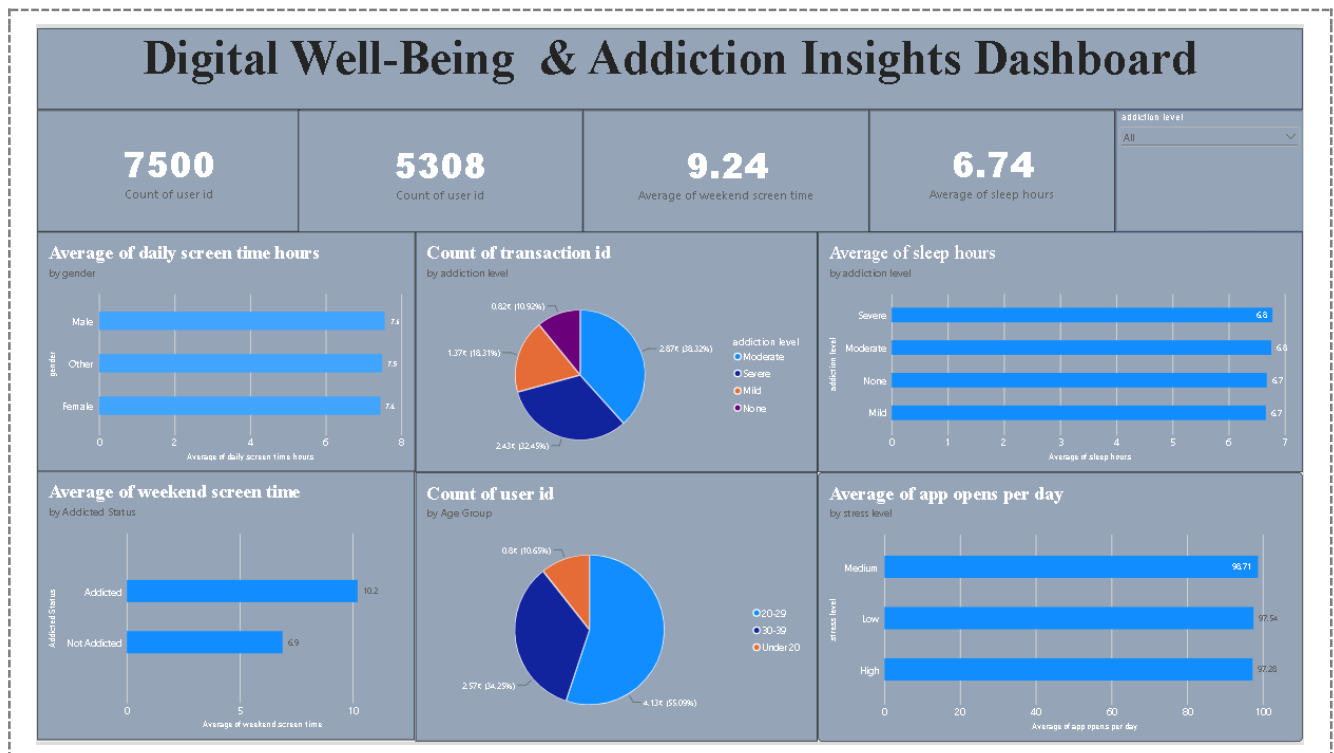


Figure 5.1: Digital Well-Being & Addiction Insights Dashboard

- **Addiction is common:** 5,308 of 7,500 users (**70.8%**) are addicted. Moderate is 38.3%, Severe 32.5%, Mild 18.3%, none 10.9%.
- **Weekends are heavier:** average weekend screen time is **9.24 h** against 7.50 h on a normal day. Addicted users average **10.21 h**, non-addicted 6.89 h.
- **Sleep is short for everyone:** average **6.74 h**; 33.0% sleep under 6 h and only 43.7% sleep 7 h or more. It barely changes with addiction level (6.66 to 6.78 h).
- **Gender, stress and age make little difference:** daily screen time is 7.45 to 7.56 h across genders and app opens are 97 to 99 a day across stress levels. The 20-29 group is the largest (about 55% of users).

6. Report

6.1. Report Design Screenshot with Insights

Report 1 – Digital Addiction & Screen Time Behaviour Analysis has cards (app opens, social media hours, notifications), addiction-level and Age Group slicers, bar charts of users by addiction level, work/study hours by academic impact and weekend screen time by addiction level, a line chart of weekend screen time by age group, and a Decomposition Tree (users by Age Group, gender, stress, addiction level, impact).

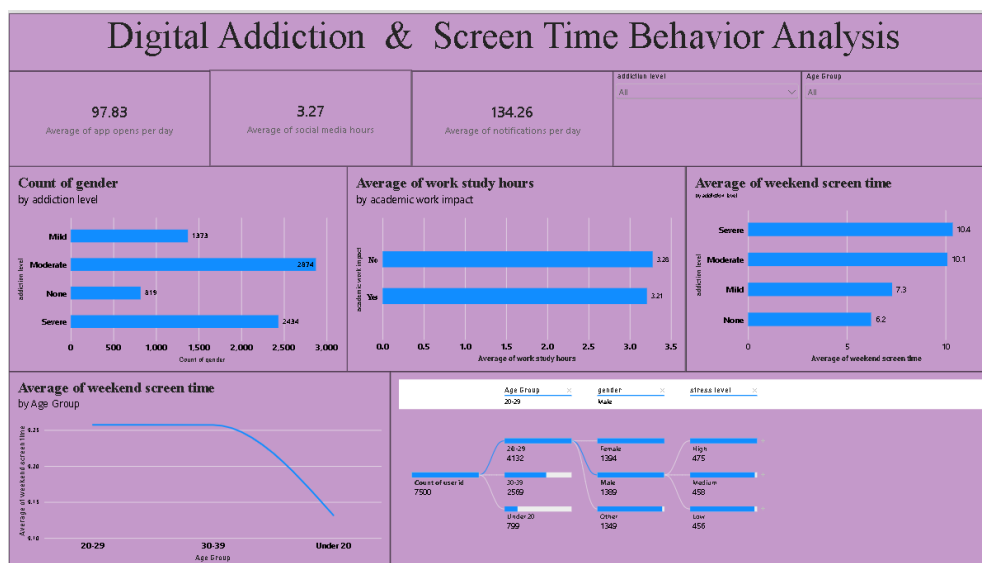
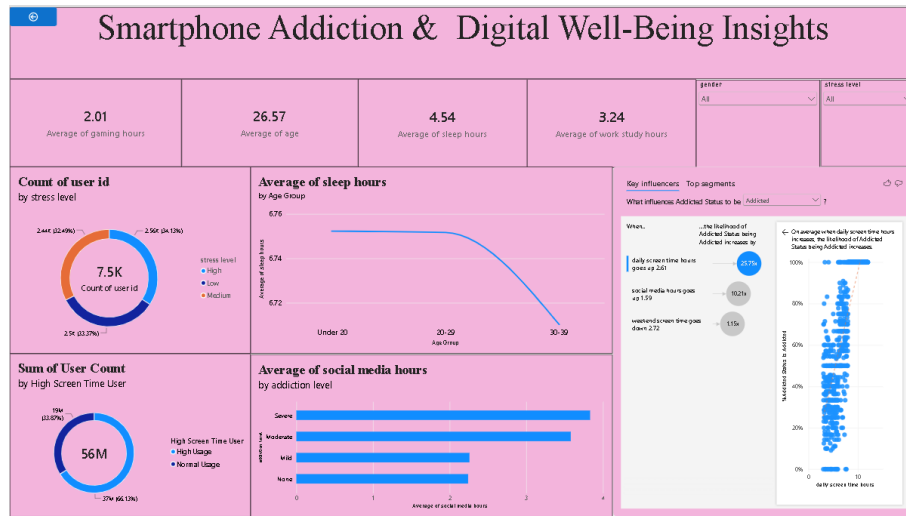


Figure 6.1: Digital Addiction & Screen Time Behavior Analysis

- **Report 1:** users open apps **97.8 times** and get **134 notifications** a day. Weekend screen time rises with addiction level: 6.23 h (none), 7.29 h (Mild), 10.09 h (Moderate), 10.36 h (Severe). Weekend use is almost the same across age groups (9.1 to 9.3 h).
- **Report 1:** users who report an academic/work impact study or work 3.21 h a day against 3.28 h for others, and the split is exactly even (3,747 vs 3,753), so impact is not explained by hours.

Report 2 – Smartphone Addiction & Digital Well-Being Insights has cards (gaming, age, sleep, work/study), gender and stress slicers, donuts for stress level and High Screen Time User, a line chart of sleep by age group, a bar chart of social media by addiction level, a **Key Influencers** visual (target: Addicted Status) and a Back button.



- **Report 2:** stress is evenly spread (High 34.1%, Low 33.4%, Medium 32.5%) and addicted and non-addicted users have the same mix. Social media time grows with addiction: 2.26 h (Mild), 3.58 h (Moderate), 3.83 h (Severe).
- **Report 2: daily screen time is the strongest driver of addiction** (correlation with the addicted label $r = 0.58$), then weekend screen time (0.56) and social media hours (0.41). Gaming, notifications, app opens and age show almost no relationship ($|r| \leq 0.01$).

7. Performance Testing

The model is lightweight: one table of 7,500 rows and 16 source columns (the .pbix file is about 0.3 MB).

7.1. Utilization of Data Filters

Filter type	Field	Where used
Slicers (8)	gender (2), Age Group (2), addiction level (3), stress level (1)	Visualizations, Dashboard, Report 1, Report 2
Visual-level filters (3)	Addicted Status = Addicted (2); sleep hours Bottom 10 (1)	KPI card, Dashboard card; Report 2 sleep card
Cross-filtering	All visuals on a page	All pages

7.2. Number of Calculation Columns / Measures

4 calculated columns (Age Group, Addicted Status, High Screen Time User, User Count) and **0 explicit DAX measures**; the visuals use built-in Average, Distinct Count, Count and Sum aggregations, Don't Summarize.

7.3. Number of Visualizations

Page	Cards	Charts	Slicers	AI visuals	Other	Total
KPI	11	0	0	0	0	11
Visualizations	0	6	3	0	0	9
Dashboard	4	6	1	0	1 (title)	12
Report 1	3	4	2	1	1 (title)	11
Report 2	4	4	2	1	2 (title, button)	13
Total	22	20	8	2	4	56

8. Conclusion / Observations

1. **70.8% of users are addicted** (Moderate or Severe); Moderate is the largest group (38.3%).
2. **Total screen time is the clearest marker of addiction:** addicted users average 8.47 h a day against 5.16 h, and no non-addicted user exceeds 8 hours.
3. **Social media, not gaming, separates the groups:** 3.70 h for addicted users against 2.25 h, while gaming is about 2.0 h for both.
4. **Weekends amplify the problem:** about 10.2 h per weekend day for addicted users against 6.9 h.
5. **Sleep is short for everyone** (6.74 h) and does not differ by addiction level, gender or age group; stress, gender, age, notifications and app opens do not separate the groups either.

Daily screen time	Users	Addicted	Avg sleep (h)
Up to 5 hours	1,683	36.2%	6.70
5 to 7 hours	1,626	48.5%	6.74
7 to 9 hours	1,665	83.1%	6.70
Above 9 hours	2,526	100.0%	6.78

Screen-time limits, app timers and weekend screen-free windows, starting with social media, are therefore the most promising interventions. **Limitations:** the findings are correlations from a single snapshot; most attributes are almost evenly distributed and addiction is tightly tied to screen time, which suggests the data may be synthetic, so results should be validated on real survey data; there is no date field for trend analysis.

9. Future Scope

- Add a **date/time dimension** to track usage, sleep and addiction as trends, and **app-wise usage** (social, games, education).
- Build a **predictive model** (logistic regression or decision tree) for addiction risk.
- Use **real survey or device data** with a live database connection and scheduled refresh.
- Publish to the Power BI Service with **row-level security**, add **what-if parameters** for screen-time limits and a mobile layout.

10. Appendix

10.1. Source Code (if any)

```
create database smartphoe_usage;
use smartphone;
select * from smartphone_usage;
```

-- Gender level addiction counts

```
SELECT gender, COUNT(*), SUM(CASE WHEN addicted_label = 1 THEN 1 ELSE 0
END) AS addicted_count
FROM smartphone_usage
GROUP BY gender;
```

-- Baseline usage averages

```
SELECT AVG(daily_screen_time_hours), AVG(sleep_hours), AVG(stress_level = 'High')
FROM smartphone_usage;
```

-- Null/mising-value check

```
SELECT * FROM smartphone_usage
WHERE age IS NULL OR daily_screen_time_hours IS NULL OR addiction_level IS
NULL;
```

-- Stress level comparison by addiction status

```
SELECT addicted_label, stress_level, COUNT(*)
FROM smartphone_usage
GROUP BY addicted_label, stress_level
ORDER BY addicted_label, stress_level;
```

-- Addiction level distribution

```
SELECT addiction_level, COUNT(*), AVG(daily_screen_time_hours), AVG(sleep_hours)
FROM smartphone_usage
GROUP BY addiction_level;
```

-- Screen time comparison by academic/work impact

```
SELECT academic_work_impact, AVG(daily_screen_time_hours),
AVG(social_media_hours), AVG(gaming_hours)
FROM smartphone_usage
GROUP BY academic_work_impact;
```

10.2. GitHub & Project Demo Link

Item	Link
GitHub repository	https://github.com/anandasowjanya05/Smartphone-Usage-Digital-Addiction-Well-Being-Analytics
Project demo (video / Power BI Service link)	https://drive.google.com/file/d/1waa2J2pd3aiurlvEcsCG2QRpq0rrN2xV/view?usp=sharing

